

REST API Reference Guide

Request Payloads & HTTP Responses

Author: Shaik Mahammad Shariff

REST API Documentation

This document describes the routing endpoints, request parameters, JSON payloads, and response structures for the CreditGuard AI scoring server.

1. Web Views Routing

Route	Method	Description
`/`	`GET`	Landing home page of the Credit Card platform.
`/about`	`GET`	General project overview, business definition, and motivation.
`/predict`	`GET`	Application form interface (WTForms fields).
`/predict`	`POST`	Processes form inputs, runs model scoring, and renders results.
`/history`	`GET`	Dashboard displaying the log of last 50 decisions.

2. Health Monitoring API

```
#### `GET /health`
```

Returns the status, version, loaded model, and timestamp.

Response Example (JSON - 200 OK):

```
{
  "status": "healthy",
  "version": "1.0.0",
  "model_loaded": "logistic_regression",
  "timestamp": "2026-06-29 17:45:00"
}
```

3. Scoring REST API

```
#### `POST /api/predict`
```

Calculates credit delinquency risk for a single applicant record.

```
##### Request Headers:
```

- `Content-Type: application/json`

Request Payload Example:

```
{
  "code_gender": "M",
```

```
"cnt_children": 0,  
"cnt_fam_members": 2,  
"age_years": 35.0,  
"amt_income_total": 150000.0,  
"flag_own_car": "Y",  
"flag_own_realty": "Y",  
"name_income_type": "Working",  
"name_education_type": "Higher education",  
"name_family_status": "Married",  
"name_housing_type": "House / apartment",  
"years_employed": 5.5,  
"flag_unemployed": 0  
}
```

Response Example (JSON - 200 OK):

```
{  
  "decision": "Approved",  
  "approval_probability_percent": 98.54  
}
```

Error Response Example (JSON - 400 Bad Request):

```
{  
  "error": "Age must be between 18 and 120 years."  
}
```